

feelzlike: US and Canada snow elevation audit

Audited 24 Sep 2026 against the live site (feelzlike.com, `bundle index-C9prnJ5K.js`), the live weather API, and the GitHub repo `DarkHorse263/Yamanouchi-Weather-App` (last commit 7 Sep 2026). Map-pin terrain heights come from the Open-Meteo elevation API.

Summary

374 US and Canadian mountain pages checked. Every weather request returned 200, so none of them are crashing. The problem is the height the snow forecast is calculated for.

Group	Mountains	What's wrong	Typical error
A: base entered as summit	30 (US)	The value stored as the summit is really the base	Snow forecast about 285 m below the base
B: forecast height = base	57 (US)	The western-US catalogue uses the base as the forecast height	Snow forecast 300 m below the base
C: mid-mountain calc drops below base	18 (17 CA, 1 US)	The calculation lands just below a known base	Usually 10–40 m below base
D: no base or summit data	142 (US, East and Midwest)	Height is the map pin's ground level, then another 15% is subtracted	Below the pin, which is usually near the base lodge
OK	127 (73 US, 54 CA)	Summit used correctly	—

Only about a third of US and Canadian mountains get a snow forecast at a sensible on-mountain height.

Root cause

`midMountainElevation(summitM)` in `artifacts/feelzlike/src/lib/elevation.ts` assumes `elevationM` is the summit and subtracts up to 300 m. But `elevationM` holds different things in different places:

- In authored region files (such as `jackson-hole.ts`) it's often the base or village height.
- In the western-US catalogue, `forecastElevationM` is set equal to `baseElevationM`.
- In the East/Midwest catalogue, it's the map-pin ground height.

The server (`api-server/src/routes/weather.ts`) stores the true summit for the authored resorts, so the two sides disagree. For example, Jackson Hole is 1,924 m in the client and 3,185 m on the server.

Correction to my earlier review: Jackson Hole requests 1,635 m, not 163 m. My earlier readout was cut off. It is still wrong, since it's 289 m below Teton Village.

Worst flagship examples (group A)

Resort	Snow forecast height now	Base	Summit (server)	Suggested (midpoint)
Heavenly	1,621 m (below Lake Tahoe's surface)	1,907	3,068	2,488
Palisades Tahoe	1,606 m	1,890	2,758	2,324
Mammoth	2,123 m	2,423	3,369	2,896
Alta	2,300 m	2,600	3,374	2,987
Breckenridge	2,626 m	2,926	3,914	3,420
Park City	1,773 m	2,100 (verified)	3,056	2,578
Jackson Hole	1,635 m	1,924	3,185	2,555
Stowe	347 m	408	1,340	874
Killington	302 m	355	1,293	824

Correct examples for comparison: Big Sky, Telluride, Revelstoke and Banff Sunshine all use the summit correctly.

Recommended fix

1. Make the fields explicit in MountainLink: `baseElevationM` and `summitElevationM`. Retire the ambiguous `elevationM`.
2. Replace the snow height calculation with one that never goes below the base:

```
export function snowForecastElevation(baseM?: number, summitM?: number, pinM?: number) {
  if (baseM && summitM && summitM > baseM) return Math.round(baseM + (summitM - baseM) / 2);
  if (summitM) return Math.max(baseM ?? 0, midMountainElevation(summitM));
  return baseM ?? pinM; // never subtract from a base or pin height
}
```

3. Data:

- Group A: copy the summit from `weather.ts` into each authored mountain and keep the current value as the base.
 - Group B: the western-US catalogue already has `topElevationM`, so the new function fixes these automatically.
 - Group D: source base and summit heights. Until then, use the pin height as-is instead of subtracting from it.
4. Add a test that fails if any snow forecast height is below the base, or if the client and server elevations differ by more than 150 m.

Also noted

- Thredbo still returns 500 (the cloud-cover decimal validation error) as of this audit.
- Worth checking by hand: the Powder King base (catalogue 1,267 m, map pin 963 m), and Mt Eyak and Hilltop listed with bases of 15 m and 75 m.

The full per-mountain list is in `us_canada_snow_elevation_audit.csv`.